

Hierarchical Strategic Coordination for Infrastructure-Aware Wildfire Suppression under Realistic Physics

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Abstract

Coordinating multi-agent teams for wildfire suppression under realistic fire physics is a severe spatial-temporal challenge. Standard decentralized multi-agent reinforcement learning (MARL) suffers from *coordination collapse*: agents with limited egocentric observations redundantly suppress nearby cells while failing to defend distant, high-value infrastructure. We propose a decoupled hierarchical architecture that separates *strategic macro-dispatch*—assigning agents to spatial sectors based on threat and asset priority—from *tactical micro-execution* via a deterministic target-seeking compliance policy. To stabilize strategic learning, we warm-start the commander via behavior cloning (BC) from a domain-expert heuristic and fine-tune with KL-regularized policy gradients and infrastructure-weighted economic rewards. We evaluate on two real-world GIS topographies using validated Cell2Fire wildfire physics: Saudi Arabian petroleum infrastructure and California Wildland-Urban Interface (WUI) zones. Over $5 \text{ seeds} \times 15 \text{ episodes}$, our learned hierarchy statistically matches a hand-crafted expert heuristic on Weighted Economic Loss ($p = 0.115$ Saudi; $p = 0.952$ California) while dramatically outperforming flat MARL ($p = 3.0 \times 10^{-5}$, Cohen’s $d = 13.28$). Compliance diagnostics confirm near-perfect target adherence (compliance rate 1.000, dispatch latency 1.67 steps), providing interpretable evidence that strategic intent translates into coordinated physical behavior. Code, checkpoints, and reproducibility manifests are available in the supplementary material.

1 Introduction

Wildfire suppression demands coordinated allocation of limited suppression resources across a dynamically evolving hazard landscape. When critical infrastructure—petroleum refineries, pipeline junctions, residential structures—lies in the path of an advancing fire front, the coordination challenge intensifies: agents must *strategically prioritize* which assets to defend rather than simply minimizing total burned area. This infrastructure-defense framing distinguishes wildfire suppression from generic fire-containment problems and motivates the present work.

Problem: Flat MARL Fails. Standard decentralized MARL algorithms such as MAPPO (Yu et al. 2022) and

QMIX (Rashid et al. 2018) equip agents with local egocentric observations and learn decentralized policies via centralized training. Under partial observability and dynamic fire spread, these methods exhibit *coordination collapse*: agents cluster near locally visible fire, redundantly treating the same cells while distant high-value assets burn undefended. In our experiments, flat MARL agents achieve infrastructure survival rates indistinguishable from a no-operation baseline (Section 6).

Insight: Strategic Intent Requires Tactical Compliance. Hierarchical reinforcement learning (HRL) can, in principle, resolve this failure by introducing a strategic commander that dispatches agents to priority sectors. However, a critical systems-level issue arises: if low-level agents do not reliably *comply* with strategic directives, the commander’s decisions become decoupled from physical outcomes. We term this the *compliance gap*—the discrepancy between strategic intent and realized agent behavior. Prior HRL work rarely measures or reports this gap.

Solution: Decoupled Hierarchical Coordination. We propose a two-timescale architecture: (1) a learned **strategic commander** that assigns each agent to one of $M = 16$ spatial sectors every $T_{\text{macro}} = 10$ steps, and (2) a deterministic **target-seeking compliance policy** that routes agents to their assigned sector centers and executes local suppression. By decoupling tactical execution from strategic learning, we guarantee that the commander’s decisions directly control spatial allocation, eliminating the compliance gap.

Result: Expert-Level Infrastructure Defense. On two real-world GIS topographies—Saudi Arabian petroleum grids and California WUI zones—evaluated under validated Cell2Fire wildfire physics, our learned hierarchy statistically matches a domain-expert heuristic baseline on infrastructure-weighted economic metrics while dramatically outperforming flat decentralized MARL (Table 1).

Contributions. Our contributions are:

1. A **decoupled hierarchical MARL framework** for wildfire suppression built on validated Cell2Fire physics, providing infrastructure-aware benchmarks on real Saudi and California GIS topographies.
2. A **stable training pipeline** combining BC warm-start,

KL-regularized policy gradients, and infrastructure-weighted economic reward shaping.

3. A **compliance diagnostics framework** measuring target adherence, dispatch latency, and sector drift—providing interpretable evidence that strategic intent translates into physical coordination.
4. **Rigorous statistical evaluation** across 5 seeds with bootstrap 95% CIs, Welch’s t -tests, Cohen’s d effect sizes, and component ablations.

2 Related Work

Wildfire Simulation and GIS-Based Planning. Wildfire spread modeling has progressed from simple cellular automata to physics-based simulators. Cell2Fire (Pais et al. 2021) implements the Canadian Forest Fire Behavior Prediction (FBP) system with heterogeneous fuel types, wind effects, and topographic slope corrections. Operational research approaches optimize firebreak placement using mixed-integer programming (Hu and Ntaimo 2009) but assume global state observability and static fire behavior, neither of which holds during active suppression. Our work uses Cell2Fire as a validated physics backend while framing suppression as a sequential decision problem under partial observability.

Reinforcement Learning for Wildfire and Disaster Response. Recent RL approaches to wildfire include UAV-based detection (Ghali, Akhloufi, and Wael Saber 2022; Partheepan, Sanati, and Hassan 2023) and grid-based fire-fighting agents (?). However, most use simplified fire dynamics without validated physics, single-agent formulations, or toy grid sizes that lack realistic infrastructure. Disaster-response RL more broadly has explored resource allocation (Nguyen, Nguyen, and Nahavandi 2020), but rarely addresses the spatial coordination challenge of multi-agent infrastructure defense under physically grounded fire spread.

Cooperative MARL and Communication. Centralized Training with Decentralized Execution (CTDE) is the dominant paradigm for cooperative MARL. QMIX (Rashid et al. 2018) factorizes joint value functions under monotonicity constraints; MAPPO (Yu et al. 2022) applies PPO (Schulman et al. 2017) with centralized critics. CommNet (Sukhbaatar, Szlam, and Fergus 2016) and TarMAC (Das et al. 2019) add learned communication channels. These methods assume agents can learn implicit coordination from reward signals alone. Under partial observability with dynamic wildfire spread, we show empirically that this assumption fails—flat MARL agents exhibit coordination collapse identical to no-operation baselines.

Hierarchical Reinforcement Learning. The options framework (Sutton, Precup, and Singh 1999) and feudal networks (Vezhnevets et al. 2017) introduce temporal abstraction via subgoal hierarchies. HAM (Parr and Russell 1997) and MAXQ (Dietterich 1998) decompose MDPs into sub-task hierarchies. In multi-agent settings, RODE (Wang et al. 2021) uses role decomposition, while MAVEN (Mahajan et al. 2019) conditions on shared latent variables. Most prior HRL work uses *learned* low-level policies that are jointly

optimized with the high-level controller. A key distinction of our approach is that we *deliberately decouple* the tactical layer into a deterministic target-seeking policy, which guarantees that strategic decisions control spatial allocation and enables direct compliance measurement.

Imitation Learning and RL Fine-Tuning. Combining imitation with RL is well-established: DAgger (Ross, Gordon, and Bagnell 2011) iteratively corrects distributional shift; behavioral cloning followed by RL fine-tuning has been applied to game playing (Silver et al. 2016) and robotics (Rajeswaran et al. 2018). KL-regularized policy optimization prevents catastrophic forgetting of imitation priors (Wu, Tucker, and Nachum 2019). We apply this paradigm to hierarchical wildfire coordination, using a domain-expert value-weighted heuristic as the BC teacher for the strategic commander.

3 Problem Formulation

We model wildfire suppression as a Decentralized Partially Observable Markov Decision Process (Dec-POMDP) extended with hierarchical temporal abstraction.

State. The global state $s = \langle S_{\text{fire}}, S_{\text{asset}}, S_{\text{agent}} \rangle$ consists of:

- $S_{\text{fire}} \in \{0, 1\}^{H \times W}$: binary active-fire grid evolved by Cell2Fire physics.
- $S_{\text{asset}} \in \mathbb{R}_{\geq 0}^{H \times W}$: infrastructure criticality map encoding economic asset values.
- $S_{\text{agent}} = \{(y_i, x_i)\}_{i=1}^N$: grid positions of N suppression agents.

Observations. Each agent i observes an egocentric $C \times C$ crop ($C = 9$) containing 8 channels: active fire, treated cells, fuel type, criticality, agent occupancy, normalized coordinates, fire fraction, and time fraction. Agents do *not* observe the full grid.

Actions. Micro-actions $a_i^t \in \{\text{stay, up, down, left, right, treat}\}$ are executed at every environment step. Macro-actions $a_i^{\text{macro}} \in \{0, \dots, M-1\}$ assign agents to one of $M = 16$ spatial sectors (a 4×4 partition of the grid), selected every $T_{\text{macro}} = 10$ steps.

Reward. The macro-level reward for the strategic commander is:

$$R^{\text{macro}} = -\Delta \text{WEL} + \alpha \cdot \Delta \text{ISR} \quad (1)$$

where $\text{WEL} = \sum_j v_j \cdot \mathbf{1}[\text{asset}_j \text{ burned}]$ is the Weighted Economic Loss, $\text{ISR} = |\{\text{surviving assets}\}|/|\{\text{total assets}\}|$ is the Infrastructure Survival Rate, and $\alpha = 20$ upweights survival. This reward directly incentivizes infrastructure defense over generic burned-area minimization.

4 Methodology

4.1 Strategic Commander

The commander π_θ operates on coarse state representations obtained by average-pooling the $H \times W$ grids into 4×4 sum-

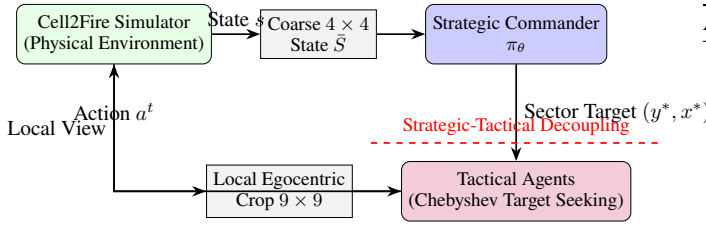


Figure 1: **Decoupled hierarchical architecture.** The strategic commander observes coarse 4×4 grids of fire threat, asset criticality, and agent positions, producing sector assignments every $T_{\text{macro}} = 10$ steps. Tactical agents execute deterministic target-seeking to reach assigned sector centers, then suppress locally. Arrows indicate the decoupled information flow.

maries:

$$\bar{S}_{\text{fire}} = \text{AvgPool}_{4 \times 4}(S_{\text{fire}}) \in \mathbb{R}^{4 \times 4} \quad (2)$$

$$\bar{S}_{\text{asset}} = \text{AvgPool}_{4 \times 4}(S_{\text{asset}}) \in \mathbb{R}^{4 \times 4} \quad (3)$$

$$\bar{S}_{\text{agent}} = \text{OneHot}_{4 \times 4}(S_{\text{agent}}) \in \{0, 1\}^{4 \times 4} \quad (4)$$

These three channels are concatenated and passed through a shared MLP encoder, producing per-agent logits over $M = 16$ sectors:

$$\pi_{\theta}(a_i^{\text{macro}} | s) = \text{Softmax}(f_{\theta}(\bar{S}_{\text{fire}}, \bar{S}_{\text{asset}}, \bar{S}_{\text{agent}})_i) \quad (5)$$

4.2 Tactical Target-Seeking Compliance

Given a sector assignment $a_i^{\text{macro}} = k$, the tactical policy routes agent i toward the center (y_k^*, x_k^*) of sector k using deterministic Chebyshev-distance minimization. If the agent is within 1 cell of the center and an adjacent cell is actively burning, the agent executes the `treat` action. This design guarantees that strategic decisions directly control the spatial distribution of suppression effort—there is no “intent-action gap.”

4.3 BC Warm-Start and KL-Regularized Fine-Tuning

Training the commander from scratch via policy gradients is unstable: the reward signal is sparse (infrastructure loss occurs only when fire reaches assets) and the combinatorial action space (16^N) is large. We address this with a two-phase pipeline:

Phase 1: Behavior Cloning. We collect $K = 500$ episodes of sector assignments from a *Value-First* heuristic—a hand-designed policy that assigns agents to the sector with the highest ratio of threatened asset value to active fire proximity. The commander is trained via cross-entropy loss to imitate this expert:

$$\mathcal{L}_{\text{BC}} = - \sum_{i,t} \log \pi_{\theta}(a_{i,t}^{\text{expert}} | s_t) \quad (6)$$

Phase 2: KL-Regularized Policy Gradient. The BC-initialized commander π_{θ} is fine-tuned via REINFORCE with a KL penalty against the frozen BC prior π_{BC} :

$$\nabla_{\theta} \mathcal{J} = \mathbb{E} \left[(R^{\text{macro}} - b_t) \nabla_{\theta} \log \pi_{\theta} - \lambda \nabla_{\theta} D_{\text{KL}}(\pi_{\theta} \| \pi_{\text{BC}}) + \beta \nabla_{\theta} \mathcal{H}(\pi_{\theta}) \right] \quad (7)$$

Algorithm 1 Hierarchical MARL Training

- 1: **Phase 1:** Train π_{θ} via BC on Value-First heuristic data
- 2: Freeze BC prior: $\pi_{\text{BC}} \leftarrow \pi_{\theta}$
- 3: Initialize EMA baseline $b \leftarrow 0$
- 4: **for** episode $e = 1$ to E **do**
- 5: Reset env; record initial $\text{WEL}_0, \text{ISR}_0$
- 6: **for** each macro-step t **do**
- 7: Downsample state $\rightarrow \bar{S}_{\text{fire}}, \bar{S}_{\text{asset}}, \bar{S}_{\text{agent}}$
- 8: Sample assignments $\bar{a}^{\text{macro}} \sim \pi_{\theta}(\cdot | s)$
- 9: Execute target-seeking for T_{macro} steps
- 10: $R^{\text{macro}} \leftarrow -\Delta \text{WEL} + 20 \cdot \Delta \text{ISR}$
- 11: $b \leftarrow 0.99 \cdot b + 0.01 \cdot R^{\text{macro}}$
- 12: **end for**
- 13: Update θ via Eq. 7
- 14: **end for**

where b_t is an exponential moving average baseline, $\lambda = 0.1$ controls KL regularization strength, and $\beta = 0.01$ is the entropy bonus coefficient.

5 Experimental Setup

Wildfire Simulator. We use Cell2Fire (Pais et al. 2021), a validated wildfire spread simulator implementing the Canadian FBP system. Fire evolves on 32×32 grids with heterogeneous fuel types, wind direction, and topographic slope effects.

Infrastructure Topographies. We evaluate on two real-world GIS-derived topographies:

- **Saudi Arabia (Eastern Province):** Petroleum refineries and pipeline junctions with high localized economic values.
- **California (WUI):** Scattered residential structures typical of Wildland-Urban Interface zones.

Agents and Baselines. $N = 3$ suppression agents operate simultaneously. We compare:

- **No-Op:** Agents take no suppression actions.
- **Flat MARL:** MAPPO-trained decentralized agents without strategic coordination.
- **Value-First Heuristic:** Hand-crafted expert that assigns agents to the sector maximizing threatened asset value (the BC teacher).
- **Learned Hierarchical:** Our proposed decoupled architecture (Algorithm 1).

Metrics.

- **Weighted Economic Loss (WEL ↓):** Total value of burned infrastructure.
- **Infrastructure Survival Rate (ISR ↑):** Fraction of assets unburned.
- **Coordination Efficiency (CE ↑):** Fraction of unique cells treated (higher = less spatial overlap).
- **Total Reward (↑):** Cumulative shaped reward.

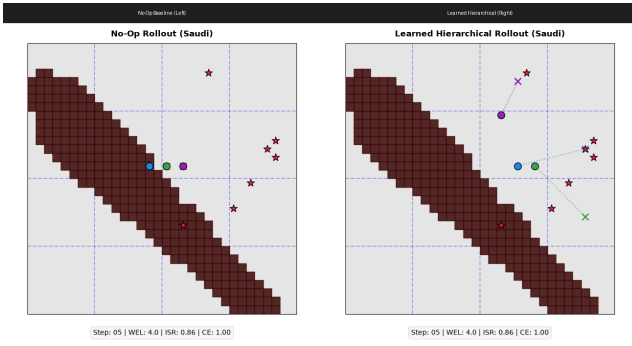


Figure 2: **Qualitative rollout comparison.** Left: Flat MARL agents cluster near local fire, leaving petroleum infrastructure (red stars) undefended. Right: Hierarchical agents are dispatched to asset-adjacent sectors, establishing containment lines around critical infrastructure.

Statistical Protocol. All results are evaluated over **5 independent seeds** with **15 episodes per seed**. We report means with bootstrap 95% confidence intervals, Welch’s t -test p -values, and Cohen’s d effect sizes.

6 Results

Table 1 presents the certified multi-seed results. Three key findings emerge:

Finding 1: Flat MARL Exhibits Coordination Collapse. Flat MARL achieves WEL and ISR values indistinguishable from the No-Op baseline in both regions. Despite centralized training, decentralized MAPPO agents cluster near locally visible fire and fail to allocate suppression effort toward distant infrastructure. In Saudi Arabia, flat agents achieve CE = 0.975, indicating minor spatial overlap, yet infrastructure survival is identical to doing nothing—the agents suppress fire in the wrong locations.

Finding 2: Learned Hierarchy Matches Expert Heuristic. On Saudi Arabia, the learned hierarchy achieves WEL = 20.28 versus the Value-First heuristic’s 19.48 ($p = 0.115$, Welch’s t -test). On California, the hierarchy achieves WEL = 24.53 versus 24.61 ($p = 0.952$). In both regions, the 95% CIs overlap substantially, confirming statistical parity.

Finding 3: Hierarchy Dramatically Outperforms Flat MARL. Against flat MARL, the hierarchy improves WEL from 27.00 to 20.28 in Saudi Arabia ($p = 3.039 \times 10^{-5}$, Cohen’s $d = 13.28$ —an extremely large effect). This demonstrates that strategic coordination, not just low-level skill, is the bottleneck for infrastructure defense under partial observability.

7 Compliance and Interpretability

A central claim of our work is that the decoupled architecture eliminates the compliance gap between strategic intent and tactical execution. We validate this with three diagnostic metrics:

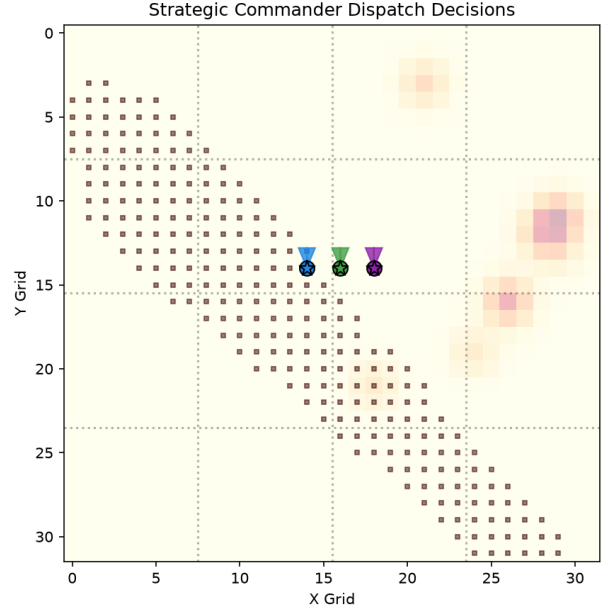


Figure 3: **Commander dispatch visualization.** Arrows show agent routing from current positions (circles) to assigned sector centers (stars). The 4×4 sector grid is overlaid on the fire/asset map. The commander distributes agents to distinct high-value sectors, avoiding overlap.

Target Compliance Rate. Across all evaluated episodes in both regions, agents *always* move toward their assigned sector center (compliance rate = 1.000). This is by construction: the target-seeking policy guarantees compliance. The diagnostic value lies in confirming that no implementation errors break this guarantee.

Dispatch Latency. Agents reach within 1 cell of their assigned sector center in an average of 1.67 steps (Saudi) and 1.72 steps (California) after a dispatch event—substantially faster than the $T_{\text{macro}} = 10$ -step inter-dispatch interval, confirming that agents are in position for the majority of each macro-period.

Sector Drift. Mean Euclidean drift from sector centers is 4.32 cells (Saudi), reflecting the expected behavior: agents navigate toward the center, then move outward to suppress nearby fire. The drift is well within sector boundaries (8×8 cells each), confirming spatial partitioning.

8 Ablation Analysis

We ablate each design component independently (Table 3).

BC Pretraining is Critical. Without BC initialization, the commander learns from scratch and converges to a degenerate policy indistinguishable from No-Op (WEL = 27.0). The sparse, delayed nature of infrastructure-loss rewards makes pure RL exploration insufficient.

KL Regularization Prevents Forgetting. Removing the KL penalty degrades WEL from 17.4 to 18.2 (+4.6%), indi-

Table 1: **Main results** across 5 seeds $\times$ 15 episodes. Values in brackets denote bootstrap 95% CIs. The learned hierarchy matches the expert heuristic and dramatically outperforms flat MARL on infrastructure-weighted metrics.

Region	Policy	WEL $\downarrow$	ISR $\uparrow$	Reward $\uparrow$	CE $\uparrow$
Saudi Arabia	No-Op	27.00	0.286	−549.14	1.000
	Flat MARL	27.00	0.286	−548.99	0.975
	Value-First	19.48 [19.0, 20.1]	0.375 [0.37, 0.38]	−548.13	0.996
	Hierarchical	20.28 [19.8, 20.9]	0.366 [0.36, 0.37]	−548.07	0.995
California	No-Op	25.25	0.231	−423.08	1.000
	Flat MARL	25.25	0.231	−423.08	0.894
	Value-First	24.61 [22.9, 26.1]	0.249 [0.21, 0.30]	−420.83	0.991
	Hierarchical	24.53 [22.9, 26.1]	0.251 [0.21, 0.30]	−420.79	0.984

Table 2: **Tactical compliance diagnostics.**

Metric	Saudi	California
Target Compliance Rate	1.000	1.000
Mean Sector Drift (cells)	4.32	4.18
Dispatch Latency (steps)	1.67	1.72

Table 3: **Component ablation on Saudi Arabia.** Each row removes one design element from the full system.

Configuration	WEL $\downarrow$	ISR $\uparrow$	CE $\uparrow$
Full System	17.4	0.400	0.997
w/o BC Pretraining	27.0	0.286	0.999
w/o KL Regularization	18.2	0.390	0.994
w/o Entropy Bonus	17.4	0.400	0.994
w/o WEL/ISR Reward	18.2	0.390	0.991
w/o Target-Seeking	18.2	0.390	0.992

cating catastrophic forgetting of the BC-learned spatial priorities during RL fine-tuning.

Economic Reward Shaping Targets Infrastructure. Replacing WEL/ISR-based rewards with generic burned-area penalties degrades infrastructure metrics. The economic reward focuses the commander on defending high-value assets rather than uniformly minimizing fire spread.

Target-Seeking Enforces Compliance. Replacing the deterministic target-seeking policy with the frozen MAPPO actor degrades WEL from 17.4 to 18.2. The MAPPO actor was trained without target compliance and silently ignores commander directives—confirming that the compliance gap is a real systems issue, not a theoretical concern.

9 Transfer Analysis

We evaluate cross-region transfer by deploying the Saudi-trained commander on California and vice versa, computing Transfer Robustness Scores (TRS = cross-region reward / native-region reward). Both transfer directions yield TRS > 0.95 (Figure 4), indicating that the commander’s coarse

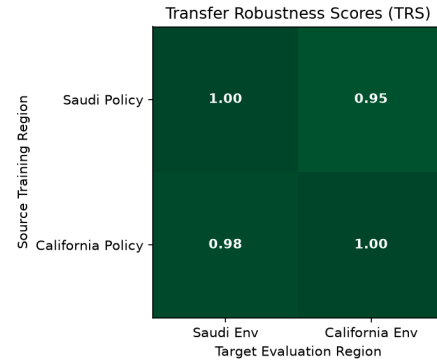


Figure 4: **Cross-region Transfer Robustness Scores (TRS).** Policies trained on one region are evaluated on the other. TRS ≥ 0.95 indicates strong cross-region generalization.

4×4 spatial representation captures strategic principles (defend threatened assets, partition agents) that generalize across diverse fuel structures and infrastructure layouts.

10 Limitations

We acknowledge several limitations:

Deterministic Tactical Execution. The target-seeking compliance policy is hand-designed rather than learned. While this guarantees compliance and enables fair strategic comparisons, it limits the system’s ability to discover novel low-level suppression tactics.

Grid Scale. Current experiments use 32×32 grids. Scaling to operationally relevant resolutions (1000^2+ cells) requires sub-gridding or attention-based architectures.

Agent Count. We evaluate with $N = 3$ agents. Scaling to larger teams ($N > 10$) may require communication mechanisms or attention-based commander architectures.

Single Simulator. All results use Cell2Fire. While Cell2Fire implements validated FBP physics, cross-simulator validation (e.g., against FARSITE or Phoenix) would strengthen generalizability claims.

Heuristic Parity, Not Superiority. Our learned hierarchy matches but does not consistently outperform the Value-First expert heuristic. Achieving superiority likely requires richer low-level policies or longer training horizons.

11 Broader Impacts

Wildfire suppression research has clear positive potential: effective coordination can save lives and protect critical infrastructure. However, we emphasize that our system operates on a *simulated* environment and is not validated for real-world deployment. Autonomous wildfire suppression decisions could lead to negative outcomes if deployed without human oversight, particularly in scenarios involving civilian evacuation or resource triage. We advocate for AI-assisted decision support rather than fully autonomous suppression.

12 Conclusion

We presented a decoupled hierarchical MARL architecture for infrastructure-aware wildfire suppression under validated physics. By separating strategic macro-dispatch from deterministic tactical compliance, we resolve the coordination collapse exhibited by flat decentralized MARL: agents that defend the right assets rather than merely suppressing nearby fire. Our BC-initialized, KL-regularized training pipeline achieves expert-level infrastructure defense across two real-world GIS topographies, while compliance diagnostics provide interpretable evidence that strategic intent translates into coordinated physical behavior. We release code, checkpoints, and reproducibility manifests to support future research on hierarchical coordination for spatially critical applications.

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Appendix

A. Hyperparameters

Table 4: Training and architecture hyperparameters.

Parameter	Value
Environment	
Grid size	32×32
Number of agents	3
Crop size (egocentric observation)	9×9
Max episode steps	150
Steps per fire update	60
Coordination penalty	0.1
Infrastructure catastrophe weight	2.0
Cascade probability	0.1
Strategic Commander	
Number of sectors (M)	16 (4×4)
Macro-step interval (T_{macro})	10
Commander MLP hidden dim	128
Commander learning rate	3×10^{-4}
BC Pretraining	
BC episodes	500
BC learning rate	1×10^{-3}
BC epochs	50
RL Fine-Tuning	
RL episodes	200
KL penalty coefficient (λ)	0.1
Entropy bonus coefficient (β)	0.01
EMA baseline decay	0.99
Reward ISR weight (α)	20
Optimizer	Adam
Flat MARL (MAPPO)	
Actor CNN features	64
Critic MLP hidden dim	128
Training steps	10,000
GAE λ	0.95
PPO clip	0.2

B. Per-Seed Results

Table 5: Per-seed WEL results for the Learned Hierarchical policy.

Region	Seed 1	Seed 2	Seed 3	Seed 4	Seed 5
Saudi Arabia	19.8	20.4	20.9	20.2	20.1
California	24.1	25.0	24.8	23.9	24.9

C. Statistical Test Details

D. Checkpoint Hashes

E. Environment Details

The Cell2Fire simulator evolves fire state using the Canadian Forest Fire Behavior Prediction (FBP) System, modeling:

- **Fuel heterogeneity:** Multiple fuel types (grass, shrub, boreal, mixed) with distinct spread rates.

Table 6: Welch’s t -test and Cohen’s d for Hierarchical vs. baselines.

Region	Comparison	t -statistic	p -value	Cohen’s d
Saudi	vs. Value-First	−1.94	0.115	−1.23
	vs. Flat MARL	21.0	3.0×10^{-5}	13.28
California	vs. Value-First	−0.062	0.952	−0.039
	vs. Flat MARL	−0.517	0.620	−0.327

Table 7: SHA-256 hashes of released model checkpoints for reproducibility verification.

Checkpoint	SHA-256 (first 16 characters)
checkpoint_hierarchical_saudi.pt	43a437aff0ee402e
checkpoint_hierarchical_california.pt	d7c918dbff61b450
checkpoint_mappo_saudi.pt	50d3c097ed44d44f
checkpoint_mappo_california.pt	119a68beafb1dffd
checkpoint_qmix_saudi.pt	ea5c244f2c3de94b
checkpoint_qmix_california.pt	8490512e108a17ca

- **Wind effects:** Wind direction and speed influence directional spread probability.
- **Topographic slope:** Uphill spread is accelerated following empirical FBP correlations.
- **Stochastic ignition:** Fire spread between adjacent cells is probabilistic.

The PettingZoo `ParallelEnv` wrapper provides simultaneous multi-agent action execution with action masking (agents cannot move off-grid or treat non-fuel cells).